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COMPUTER

CHAPTER 22

Web Search Engines

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CHAPTER 22

Web Search Engines

Computer Awareness

IN THIS CHAPTER

- Search engines and directories
- Crawlers and indexes
- Ranking and retrieval
- Types of search engines
- Important search engines
- Search terminology

- DSA = Digital Signature.
- Hashing = One-way process.
- MD5 = 128-bit hash.
- SHA-1 = 160-bit hash.
- SHA-256 = Most commonly used SHA version.
- PKI = Manages certificates and public keys.
- Digital certificate = Issuer-signed binding of identity to a public key.

A Web Search Engine is a specialized software system or online service that helps users find information on the Internet by entering keywords or phrases.

It searches the World Wide Web (WWW) and returns relevant results such as:

- Web pages
- Images
- Videos
- Documents
- News articles
- Other online resources

The results displayed by a search engine are called Search Results or Hits. Example

If a user searches:

"How to bake a cake?"

The search engine displays a list of websites containing relevant information about baking a cake.

A Web Search Engine is a specialized computer server that searches for information on the World Wide Web and presents relevant results to users based on their queries.

SEARCH ENGINE VS WEB DIRECTORY

Feature	Search Engine	Web Directory
Maintained By	Algorithms and Crawlers	Human Editors
Data Collection	Automatic	Manual
Speed	Fast	Slow
Examples	Google, Bing	Yahoo Directory, DMOZ
Updating	Automatic	Manual

Exam Point

Search Engines = Automatic

Web Directories = Human Managed

SEARCH ENGINES AS DATA MINING APPLICATIONS

Modern search engines are among the largest data mining systems in the world.

They use data mining techniques in:

1. Crawling

Deciding:

- Which websites to visit
- How often pages should be revisited

2. Indexing

- Which pages should be stored
- How much information should be stored

3. Searching and Ranking

Deciding:

- Which pages are most relevant
- Which page should appear first
- Which advertisements should be displayed

Exam Point

Search engines use Data Mining, Artificial Intelligence (AI), and Machine Learning (ML) techniques to improve search quality.

CLOUD COMPUTING AND SEARCH ENGINES

Modern search engines process enormous amounts of data.

Therefore, they use Computer Clouds (Cloud Computing) consisting of thousands or even hundreds of thousands of computers working together.

Example

Google operates massive data centers worldwide to process billions of searches every day.

CRAWLERS (SPIDERS)

Search engines use special programs called:

- Crawlers
- Spiders
- Bots
- Web Robots

These programs automatically browse websites and collect information.

Example

Google's crawler is called: Googlebot

COMPONENTS OF A SEARCH ENGINE

There are four basic components of a search engine.

1. Web Crawler

A Web Crawler is a software program that automatically browses websites and collects information from web pages.

Other Names

- Spider
- Search Engine Bot
- Web Robot
- Crawler

Functions

- Visits websites
- Reads web pages
- Downloads information
- Finds new pages through links

Example :Googlebot

Exam Point :Googlebot is the crawler used by Google. 2. Database

The database is the storage area where information collected by crawlers is stored.

Features

- Stores billions of web pages
- Stores images and documents
- Enables fast searching

Examples

- Amazon Elasticsearch Service (renamed Amazon OpenSearch Service in 2021): a managed search and analytics service.
- Splunk: a search and analytics platform, rather than a generic relational database.

Exam Point :Search engine databases are generally distributed and highly scalable.

3. Search Interface

The Search Interface is the platform through which users interact with the search engine.

Functions

- Accepts user queries
- Displays search results

Example :Google Search Homepage

google.com

Exam Point: The search interface acts as a bridge between the user and the search engine database.

4. Ranking Algorithms

Ranking algorithms determine the order in which search results appear. Purpose :To show the most relevant pages first.

Example

Google's famous ranking algorithm: PageRank

TYPES OF SEARCH ENGINES

1. CRAWLER-BASED SEARCH ENGINES

Crawler-based search engines use automated programs (crawlers) to discover and index web pages.

Examples

- Google
- Bing
- Yahoo
- Baidu
- Yandex

Advantages

✔ Fast

✔ Automatically updated

✔ Huge database

Exam Point

Most modern search engines are crawler-based.

2. HUMAN-POWERED DIRECTORIES

Human-powered directories use human editors to review and categorize websites. Examples

- Yahoo Directory (Discontinued)
- DMOZ (Open Directory Project; discontinued in 2017)

Advantages

✔ Quality-controlled listings

Disadvantages

✘ Slow updates

✘ Limited scalability

3. HYBRID SEARCH ENGINES

Hybrid search engines combine:

- Automated Crawlers
- Human Editing

Examples

Historically:

- Yahoo
- Google (limited human intervention for quality control)

Exam Point

Modern hybrid search engines are becoming increasingly crawler-based.

4. Meta Search Engines

Meta Search Engines do not maintain their own database.

Instead, they collect results from multiple search engines. Example

MetaCrawler

Advantages

✔ Wider coverage

✔ Multiple search sources

5. SPECIALTY SEARCH ENGINES

These search engines focus on specific categories.

Examples

Shopping websites search(amazon), Domain Name Search etc.

HOW SEARCH ENGINES WORK

Every search engine works through three main stages.

1. Crawling

2. Indexing

3. Ranking & Retrieval

STAGE 1: CRAWLING Crawling is the process of discovering and collecting information from web pages.

Information Collected

- Page Title
- Keywords
- Content
- Images
- Links
- Metadata

Performed By

- Crawlers
- Spiders
- Bots

Example

Googlebot visits millions of websites daily.

STAGE 2: INDEXING

Indexing is the process of organizing and storing collected information for fast retrieval.

Book Library Example

Imagine:

- Reading all books = Crawling
- Creating catalog = Indexing

Search engines do the same on a much larger scale.

Exam Point :Without indexing, search engines would have to scan the entire Internet for every search. STAGE 3: RANKING AND RETRIEVAL

Ranking determines the order of search results.

Retrieval means displaying those results to users.

Important Ranking Factors

- Relevance
- Popularity
- Quality
- Freshness
- User Experience

Example

If you search:

"Best smartphones 2026"

The most useful and trusted websites appear first.

IMPORTANT TERMS

Keyword / Key Phrase

The word or phrase entered by the user into a search engine.

Example

"Computer Network"

"Cyber Security Notes"

Exam Point

Keyword = Search Query = Key Phrase

Search Engine Results Page (SERP)

The page that displays search results after a query is entered.

Example

The page displayed after searching on Google.

Web Crawler

A software program that automatically browses and indexes websites.

Other Names

- Spider
- Bot
- Web Robot

MOST POPULAR SEARCH ENGINES IN THE WORLD

1. Google

Founded :1998

Founders

- Larry Page
- Sergey Brin

Market Share

Google accounts for a large majority of global search-engine usage; the exact share changes with date, location, device and measurement.

Exam Point

Google is the world's most widely used search engine.

2.

Microsoft Bing

Developed By :Microsoft

Launch Year :2009

3. Yahoo!

Founded By

- Jerry Yang
- David Filo

Founded :1994

4. Baidu

Country :China

Founded :2000

Exam Point :Baidu is the most popular search engine in China.

5. Yandex

Country :Russia

Founded :1997

Exam Point

Yandex is the leading search engine in Russia.

6. DuckDuckGo

Founded :2008

Special Feature :Privacy-focused search engine.

Does not track users.

7. Ask.com

Founded :1996

Special Feature :Question-answer style search.

Important Correction

Safari is NOT a Search Engine Safari is a Web Browser developed by Apple in 2003.

Examples of Browsers

- Safari
- Chrome
- Firefox
- Edge

WORLD'S FIRST SEARCH ENGINE

Archie

Founded in 1990

Created By :Alan Emtage

Exam Point :Archie is considered the world's first Internet search engine.

SEARCH ENGINE VS WEB BROWSER

Feature	Search Engine	Web Browser
Purpose	Finds Information	Opens Websites
Works On	Internet Data	Displays Web Pages
Examples	Google, Bing	Chrome, Safari
Type	Online Service	Software Application

MOST IMPORTANT FACTS FOR EXAMS

1. Googlebot is Google's web crawler.

2. PageRank is Google's famous ranking algorithm.

3. SERP stands for Search Engine Results Page.

4. Crawling is performed by spiders/bots.

5. Search engines work in three stages:

◦ Crawling

◦ Indexing

◦ Ranking & Retrieval

6. Google was founded by Larry Page and Sergey Brin.

7. Yahoo was founded by Jerry Yang and David Filo.

8. Baidu is China's leading search engine.

9. Yandex is Russia's leading search engine.

10. DuckDuckGo is privacy-focused.

11. Safari is a web browser, not a search engine.

12. Archie (1990) is considered the first Internet search engine.

13. Meta Search Engines combine results from multiple search engines.

14. Human-powered directories rely on human editors.

15. Modern search engines use cloud computing and data mining.

QUICK REVISION NOTES

- Search Engine helps users find information on the Internet.
- Search Results are called Hits.
- Googlebot is Google's crawler.
- Search engines use Cloud Computing and Data Mining.
- Four components:
- Web Crawler
- Database
- Search Interface
- Ranking Algorithm
- Search engine stages:
- Crawling
- Indexing
- Ranking & Retrieval
- SERP = Search Engine Results Page.
- PageRank = Google's ranking algorithm.
- Meta Search Engines combine results from multiple search engines.
- Baidu = China.
- Yandex = Russia.
- DuckDuckGo = Privacy-focused.
- Safari = Browser, not Search Engine.
- Archie (1990) = First Internet Search Engine.

MOBILE PHONES (CELLULAR PHONES)

A Mobile Phone (Cell Phone) is an electronic communication device used to make and receive calls using radio frequency signals (electromagnetic waves) while the user is moving within a cellular network coverage area.

Mobile phones communicate through a network of specialized base stations called Cell

Sites (Mobile Towers).

A mobile phone is an electronic device used for mobile telecommunications over a cellular network, allowing users to communicate through voice calls, messages, and internet services while moving from one location to another.

Key Features of Mobile Phones

Modern mobile phones support:

- Voice Calls
- SMS (Short Message Service)
- MMS (Multimedia Messaging Service)
- Email
- Internet Access
- Bluetooth
- Camera & Video Recording